

Module 03

Ansible Playbook

DevOps end to end services & solutions

Agenda

Ansible Architecture

- > What is Playbook?
- > Ansible Playbook configuration
- > Ansible playbook tasks
- > Ansible Notify and Handlers
- > Hands-On Lab-03, 04

What is Playbook?

Ansible Playbook is a YAML-formatted file containing a blueprint of automation tasks, serving as a step by step to do list of tasks to configure systems, deploy applications, or automate IT infrastructure. They define desired states for managed nodes, enabling repeatable, scalable, and idempotent orchestration, such as managing packages, users, or services

```
tee playbook-apache.yml > /dev/null <<EOF
- hosts: slaves
  become: yes
  gather_facts: no

  tasks:

    - name: Update packages
      apk:
        update_cache: yes

    - name: Install Apache + OpenRC
      apk:
        name:
          - apache2
          - openrc
        state: present
```

Ansible Playbook configuration

Hosts

Run this playbook on all hosts in the slaves group

Become = yes

Run tasks with root privileges

Ansible Playbook configuration

Gather_facts

Collect system info

Apk and apk update

This uses the Alpine package manager
module
refreshes package index

Ansible Playbook Task

Task

A normal action:
install package
copy config
update file

Shell

Run with shell cli

Ansible Notify and Handlers

Why this matters

Without handlers:

service restart happens every time ❌

With handlers:

restart only when needed ✅

handlers are special tasks that only run when explicitly triggered by a notify directive from another task that has reported a change. This mechanism is ideal for actions like restarting a service after its configuration file has been modified, ensuring efficiency by avoiding unnecessary restarts

```
hosts: hgdev1.tekneed.com
```

```
tasks:
```

```
- name: restart httpd
```

```
  service:
```

```
    name: httpd
```

```
    state: restarted
```

```
  notify: restart httpd
```

← calling handler

```
handlers:
```

```
- name: restart httpd
```

```
  service:
```

```
    name: httpd
```

```
    state: restarted
```

← handler

Ansible Notify and Handlers

Notify

If the task changes something, it triggers a handler

Handler

A special task that runs once at the end of the play

Hands-On – Lab 03, 04

Getting Started

In this lab, we will explore Ansible playbook.

- > Create playbook based on apache web server
- > Create playbook based on python flask server
- > Create Global var
- > Display Global var

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

